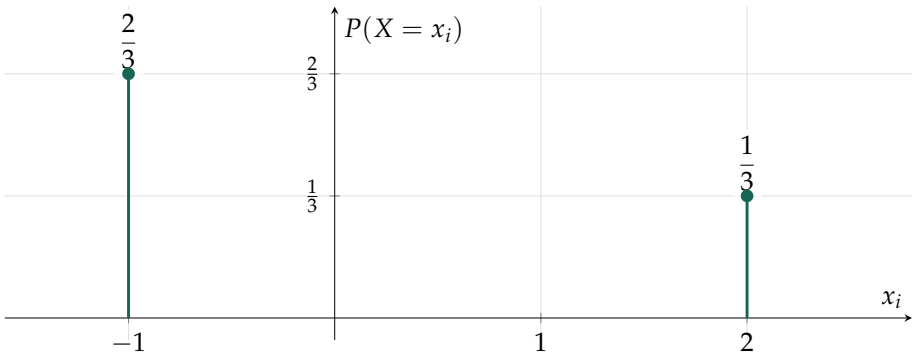
8.1 Variable, event and distribution

Key point · Definition and results. On a finite sample space Ω, a real random variable X is a function that associates a real number with each outcome. The event $\{X = a\}$ groups the outcomes giving the value a. Its distribution associates with each distinct value x_i the probability $p_i = P(X = x_i)$, with $p_i \geq 0$ and $\sum p_i = 1$.

Status and justification. Definitions. The events associated with distinct values form a partition of the sample space; this justifies the sum of the probabilities.

Example 1. A fair die gives a gain X equal to 2 for 1 or 2, and to -1 otherwise. The distribution is $P(X = 2) = \frac{1}{3}$, $P(X = -1) = \frac{2}{3}$. $\{X \leq 0\} = \{3;4;5;6\}$: probability $\frac{2}{3}$.



The height shows the probability: $P(X = -1) = \frac{2}{3}$ and $P(X = 2) = \frac{1}{3}$. The exact probabilities are written on the two bars.

Watch out. The values are not always equally likely even if the outcomes are. Group the outcomes that give the same gain.

Application. Two fair coins are tossed. X counts the heads. Write its distribution and $P(X \leq 1)$.

8.2 Expectation and linearity

Key point · Definition and results. The expectation is $E(X) = \sum_i x_i p_i$. It describes the long-term theoretical mean, not a result that is necessarily possible on a single trial. For variables X and Y on the same finite sample space and real numbers a,b: $E(aX + b) = aE(X) + b$ and $E(X + Y) = E(X) + E(Y)$, with no assumption of independence.

Status and justification. Summing over the outcomes ω, $E(X+Y) = \sum_\omega (X(\omega)+Y(\omega))P(\omega)$; distributing the sum gives the result. The constant b has expectation b because the probabilities sum to 1.

> **Example 2.** For the previous distribution $P(X=2)=\frac{1}{3}$, $P(X=-1)=\frac{2}{3}$, $E(X)=0$: the game is fair if X is the net gain. For $Y=3X-2$, $E(Y)=3\times 0-2=-2$.

> **Watch out.** An expectation is not an individual guarantee; a game can be losing at every observed horizon despite a positive expectation.

> **Application.** A gross gain is 10 with probability $\frac{1}{5}$ and 0 otherwise. What stake makes the game fair?

8.3 Variance, standard deviation and König–Huygens

> **Key point · Definition and results.** Writing $\mu = E(X)$, $V(X) = E((X-\mu)^2) = \sum_i (x_i-\mu)^2 p_i$. The standard deviation is $\sigma(X) = \sqrt{V(X)}$. The König–Huygens formula gives $V(X) = E(X^2) - E(X)^2$. A variance is always positive or zero; the standard deviation has the same unit as X, the variance a squared unit.

Status and justification. Expand $(X-\mu)^2 = X^2 - 2\mu X + \mu^2$, apply linearity and $E(X)=\mu$: $E(X^2) - 2\mu^2 + \mu^2$. We obtain the formula.

> **Example 3.** For X with values 0 and 4 with probabilities $\frac{3}{4}$ and $\frac{1}{4}$, $E(X)=1$, $E(X^2)=4$, $V(X)=3$ and $\sigma=\sqrt{3}$. Direct calculation: $(0-1)^2(\frac{3}{4}) + (4-1)^2(\frac{1}{4}) = 3$.

> **Watch out.** $E(X^2)$ is not $E(X)^2$. A variance calculated as negative signals an error or a very small rounding error to be dealt with explicitly.

> **Application.** For X with values $-1, 1$ equally likely, calculate the expectation, variance and standard deviation.

8.4 Transforming and comparing gains

> **Key point · Definition and results.** For real a,b, $V(aX+b) = a^2 V(X)$ and $\sigma(aX+b) = |a|\sigma(X)$. A translation moves the mean without changing the dispersion; a multiplication changes the scale. Two games with the same expectation can have very different dispersions.

Status and justification. $aX+b - E(aX+b) = a(X-E(X))$. Squaring and then taking the expectation gives $a^2 V(X)$. The square root of a^2 is $|a|$.

> **Example 4.** A game A always pays 2. A game B pays 0 or 4 with equal probability. Both have expectation 2; variances 0 and 4. Their distributions are therefore not interchangeable despite the same mean.

Watch out. The variance of a sum is not, in general, the sum of the variances; no rule of this type is used without hypotheses.

Application. $E(X) = 3$, $V(X) = 4$. Give the characteristics of $Y = -2X + 5$.

8.5 Samples and observed mean

Key point · Definition and results. A sample of size n is here a list of n independent realisations of the same variable X. Its observed mean m varies from one sample to another. We compare m with $\mu = E(X)$; repetitions allow us to observe the proportion of samples satisfying $|m - \mu| \leq \dfrac{2\sigma}{\sqrt{n}}$. This threshold is an experimental task of the programme, not a claim of an exact coverage rate.

Status and justification. Experimentation: neither a convergence theorem nor the normal distribution is proved here. We check the size of the lists, the calculation of the mean and the distinction between n and the number N of samples.

Example 5. For X with values 0 and 2 equally likely, $\mu = 1$, $\sigma = 1$. For n=100, threshold $\dfrac{2}{10} = 0.2$: we count the means in $[0.8; 1.2]$. A result of 956 out of 1000 means an observed frequency of 0.956 for this simulation only.

Watch out. Do not copy a single draw n times; do not announce that the proportion is necessarily 95 %.

Application. For $\mu = 5$, $\sigma = 3$ and n=36, give the experimental interval asked for.